

FLORIDA HARD ROCK · LIVING LAW · SUPERSEDES 2026.09.08-live

Sports Lock

Official Algorithm Specification

DESK_VERSION 2026.09.09-world

Status: living law. Same inputs → same ranking. No dice.

This document is the official list of **every rule** that goes into **every prediction** for **every ticket type**. Numbers, weights, and formulas below are copied from the running desk (src/lib/market/*). Where a constant is a **desk choice**, it is labeled that. Where a method is **published**, the source is named. Nothing here is invented at prediction time.

This is research. The site never places a bet. It never calls a lock. Photograph Hard Rock to lock a live number.

21+ / 1-800-GAMBLER

Picking with Intelligence Creates Confidence.

Not a lock. Photograph Hard Rock to lock the live number.
This site never places a bet.

Desk pin: DESK_VERSION 2026.09.09-world

Supersedes: 2026.09.08-live

Status: living law. Same snapshot + same seed → same ranking. No free dice.

This is research. The site never places a bet. It never calls a lock. Photograph Hard Rock to lock a live number when the Florida tape is missing or a boost is off-feed. 21+ / 1-800-GAMBLER.

This document is the official list of **every rule** that goes into **every prediction** for **every Florida Hard Rock ticket** the desk will price. Numbers, weights, and formulas are desk law unless a published method is named. Nothing is invented at prediction time. Empty feed = **Looked**, never **NO LOOK**.

Universe: **MLB, NFL, NBA, NHL, NCAAF, NCAAB** on **Hard Rock Bet Florida** only. Other sports stand down. College **player** tickets stand down in Florida; college player looks still run as **game-context only**.

How to read this

- **Look** = one noisy observation of a hidden number (chance the ticket hits, or a latent that feeds that chance).
- **Precision** = how loud that look is in the pool. Desk constant until the calibration loop overwrites it.
- **Ran / Thin / Looked** = the ticket-page stamp.
 - **Ran** — live lookup returned a real number.
 - **Thin** — live lookup, small sample. Light weight.
 - **Looked** — we fetched; feed empty. Layer is 50/50 (or the sport prior) with tiny precision. Not a skip. Not a guess. Never **NO LOOK**.
- **Prior** = Hard Rock Florida two-way no-vig, when posted. Else the sharp public two-way. Everything else is a reason to move off it, or not.
- **Snapshot** = the frozen JSON of every look + every posted line at compute time. Rankings are a pure function of snapshot + DESK_VERSION.
- **Seed** = `uint64(BLAKE2s(DESK_VERSION || snapshotId || eventId))`. Monte Carlo uses this seed. Same snapshot → same paths → same chance.
- Displayed chance is clipped at **99%**. Never 100%. Never “lock.” Never “guarantee.”

0. What a prediction is

For any ticket the desk estimates **P(this ticket hits | snapshot)**, then ranks by chance × payout × market quality × edge vs Hard Rock Florida juice.

The hidden object is no longer a single $P(\text{home wins})$. It is a **latent game state** G plus a **lineup/usage vector** U plus a **live match state** S (empty pre-tip). Every ticket — moneyline, alt spread, first basket, live remaining Ks, 8-leg custom parlay — is a function of draws from that object.

```
snapshot → looks → G, U, S
G, U, S + seed → N paths
ticket = l{path satisfies the photographed / posted proposition}
P_fair = mean(ticket)           # Monte Carlo, seeded
P_pool = invLogit(pool(looks)) # still runs, as a look
fair   = blend(P_fair, P_pool, P_market)
```

Ticket types that walk this spec (Florida Hard Rock):

- Popular: moneyline (2-way; NHL also 3-way regulation), spread / run line / puck line, game total, team total, alternate lines of all of those.

- Player tickets: legal NFL / NBA / MLB / NHL props and alt props. College athlete props = `illegal_fl` stand-down.
- Period tickets: 1st inning, F5 / first 3 / first 7, inning halves, quarter, half, NHL period, period spreads and period totals, period team totals.
- Game props: first / last / next score, first basket / first goal / first TD, race-to-X, odd/even, highest-scoring quarter, method of score, both teams to score (hockey/baseball analog: both teams to score a run/goal), overtime yes/no, exact / correct score when posted.
- Live / in-play: same families plus remaining-stat props and microbets (next play, next pitch result, current drive, next scoring play). Never The Call.
- Same-game parlays and SGP Max (multi-game builder, up to the book's cap).
- Cross-game 2 / 3 / 4+ parlays, teasers, flex / consolation parlays, round robins (priced as the underlying combo set).
- Futures and season props for the six sports.
- Boosts / enhanced-odds tickets: priced as the underlying ticket plus a juice flag. Boost is not edge.

Pipeline, in order:

1. Pull the snapshot (Hard Rock FL tape + ESPN + process feeds + crowd + weather + injuries).
2. Build **latents** `G` and **usage** `U` from the ensemble (\$4-6).
3. If the clock has started, attach **live state** `S` and condition (\$7).
4. Draw **seeded paths** and price every registered ticket from the same paths (\$8).
5. **Calibrate** displayed % toward the Florida tape when the market is thin (\$10).
6. **Rank** singles with `ticketScore`. **Propose** parlays with `parlayScore` under the high-hit ribbon law (\$11). Mood re-sorts; it does not rewrite chance.
7. Stamp **Rules the desk ran**. Same snapshot → same ranking.

1. Honesty that never moves

- Never print “lock”, “guarantee”, or 100%. Clip displayed chance at 99%.
- Delayed or aggregator numbers are not a fill. A **photographed Hard Rock Florida ticket** or a **Ran** `hardrockbet_fl two-way` is the fill.
- This site never places a bet. Paper log is a tracker, not a book.
- Florida: college **player** props illegal on Hard Rock Bet — blocked as tickets. The same looks may still inform the **game** latent.
- DraftKings / FanDuel **sportsbooks** are not licensed live Florida tickets. DK **Fantasy** is a different product.
- Do not invent Hard Rock's order book. Label every tape source.
- Do not invent a missing split, ERA, EPA, xG, minutes, last-10, or handle %. Empty = Looked.
- Kalshi and Polymarket are **research**, never a Florida fill.
- ESPN win% is a **model**, not book handle.
- Ticket count vs handle is a layer. Never copy-traded. Never auto-fade/tail the public.
- Do not scrape the Hard Rock app. Do not fake a login. Do not replay captured sessions.
- 21+ Hard Rock Bet / 18+ fantasy. 1-800-GAMBLER.

2. Math the whole desk shares

2.1 Log-odds pool

- $\text{logit}(p) = \ln(p / (1 - p))$, p clipped to $[0.015, 0.985]$.
- $\text{invLogit}(z) = 1 / (1 + e^{(-z)})$, then clip.
- Layer clips: $[0.12, 0.88]$. Final game-win clips: $[0.14, 0.86]$.
- Pool: precision-weighted mean of logits.

$$z = \sum \text{logit}(p_i) \cdot \text{precision}_i / \sum \text{precision}_i$$

$$P = \text{invLogit}(z)$$

Overdispersion shrink and James–Stein pull toward the **Florida (or sharp) market** stay as in 2026.09.08-live:

- Residual χ^2 : $\text{chi} = \sum \text{precision}_i \cdot (\text{logit}(p_i) - z)^2$, $\text{df} = n - 1$.
- Overdispersed if $\text{chi} / \text{df} > 1.35$.
- Keep market-family precision. Scale every non-market precision by $\max(0.35, 1.35 / (\text{chi}/\text{df}))$.
- Extra pull: $\lambda = \min(0.55, 0.18 + 0.5 \cdot \max(0, 1 - \max(0, 1 - \text{std}/0.14)))$, then $z \leftarrow \lambda \cdot \text{logit}(\text{market}) + (1 - \lambda) \cdot z$.

2.2 Seeded Monte Carlo

- Paths $N = 4_{000}$ default. $N = 12_{000}$ for The Call, ribbon parlays, alt tails beyond 2σ , exact-score, first-scorer, and any live micro.
- Generator: PCG64, seeded by \$0.
- No `Math.random`. No wall-clock entropy.
- Ticket fair = hits / N . Standard error $\sqrt{(p(1-p)/N)}$ is printed on the research sheet, never as the badge.

Why sim instead of one Φ . Alternate lines, first-event tickets, remaining live props, period slices, and same-game parlays are path statistics. A league- σ normal cannot emit them from one mean.

2.3 Normal CDF and Poisson

- Φ via Abramowitz & Stegun 7.1.26 when a closed form is the look (spread-implied winner, sheet cell).
- Rare counts (hits, Ks, receptions, walks, SOG, saves): Poisson or negative binomial. Desk default for overdispersed counts: **NB2** with $\phi = 1.18$ (DMP Learn 2026 extra-Poisson).
- Yards / points / PRA / rushing / receiving: normal or truncated-normal, sport σ from \$5, **scaled by game-variance latent**, not the raw league constant.
- Yes/no rare events (anytime TD, HR, stolen base, NHL goal): $1 - \exp(-\lambda)$ as the closed-form look; sim uses the underlying counting process.

2.4 Bill James log5 and Pythagorean

Unchanged. WP clipped $[0.15, 0.85]$ before log5.

Sport	Pythagorean exponent	Lineage
MLB	1.83	James / Davenport Pythagapat
NBA	13.91	Dean Oliver
NCAAB	10.25	KenPom-family scale
NHL	2.07	hockey Pythagorean
NFL / NCAAF / other	2.37	Football Outsiders / James

2.5 Recency EWMA

- Last **10** completed games (or last 10 player appearances after the minutes filter).
- $w_i \propto 0.82^i$, index 0 = latest. Blend **60% EWMA / 40% season** when both exist.
- Player minutes filter: drop games under half median minutes; if that leaves < 3 , keep raw 10.

2.6 Home-field logits

Added only when the split is season W-L, not already a home/road split.

Sport	logit bump	≈ extra win% at 50/50
MLB	0.12	~3.0 pts
NFL	0.16	~4.0 pts
NBA	0.20	~5.0 pts
NHL / NCAAF / NCAAB	0.22	~5.5 pts
other	0.14	~3.5 pts

2.7 Sport scoring priors (pace + variance)

League constants are **priors**, overwritten by the variance latent when process data Ran.

Sport	League total	Margin σ prior	Total σ prior	Pace unit
MLB	44.5	13.45	10.2	plays / EPA
NFL	54	16.2	13.5	plays / SP+
NBA	224	12.0	11.5	possessions
NCAAB	145	11.5	10.8	possessions
NHL	8.6	3.05	2.85	PA / inning
Baseball	6.1	1.85	1.65	5v5 minutes + xG

Variance latent: $\sigma \leftarrow \sigma_{\text{prior}} \cdot \text{clip}(1 + 0.65 \cdot (\text{postedTotal} / \text{league} - 1), 0.72, 1.35)$ then further scaled by weather, backup QB / backup goalie, and live chaos.

2.8 Ground truth (what “good” means)

Three numbers, in this order. The desk does not get to pick a flattering one after the fact.

- 1. Fill edge (user truth).** fair – implied(Hard Rock FL two-way) on the photographed ticket, or on hardrockbet_fl when Ran and no photo. This is what the ticket page prints as Edge.
- 2. CLV (learning truth).** Desk fair vs the last public / HR close before lock. Layers that lose CLV for 60 graded days get precision scaled by $\max(0.40, 1 - 2 \cdot \max(0, -\text{CLV_pts} / 5))$.
- 3. Calibration (honesty truth).** Among tickets whose displayed chance was in $[k, k+5)$, the hit rate must land inside a 90% Jeffreys band or the displayed formula’s quality multiplier is pulled toward the book. Brier and log-loss are the stored scores.

Grade clock: result is official league result under Hard Rock house rules (OT inclusion per market). Push = neither hit nor miss; excluded from hit rate, included in CLV as 0.

3. Live data — every rule, every time

Empty feed rule is unchanged: $p = 0.5$ (or sport prior), precision = 0.08, empty: true, thin: true, stamp **Looked**.

3.1 Florida tape (the fill)

	Live source	Stamp
two-way	The Odds API us2 / hardrockbet_fl (and labeled mirrors that name Florida)	Ran when both sides exist
one-way / boost	same feed if present; else photograph	one-way = cannot call fair two-way without a model hold
	user image of the Hard Rock ticket	Lock for the log. Overrides photo – feed feed when `
		≥ 1¢` Amer market is a SGP the fee carry
fallback	delayed ESPN scoreboard, other Odds API books	research prior only, labeled delayed

Best path on lines. Do not scrape Hard Rock. Subscribe to a licensed aggregator that exposes hardrockbet_fl. Poll pregame ~30–60s. In-play, prefer the aggregator websocket if it carries Florida; otherwise re-price off live state and wait for a photograph to lock the juice. Boosts, teaser charts, and some SGP prices will **never** be complete on an aggregator — photograph stays in the product for those.

DK/FD sportsbook numbers may sit in the pool as book looks. They are not a Florida fill.

3.2 Process feeds (world-class looks)

Every feed is attempted every snapshot. License and ToS are the operator's problem; the desk's problem is: if the file is missing, Looked.

Sport	Process look	Source family
NFL	play-by-play EPA, success rate, CPOE, xpass, series conversion, snap/route/target, pressure	nflverse / nflfastR nightly + in-week; ESPN as fallback box
NCAAF	EPA/SP+-style, explosiveness, success, returning production	cfbfastR / sportsdataverse + ESPN + published SP+/FPI when posted
NBA	pbp possessions, usage, on/off, four factors, lineup net, tracking shot quality when posted	stats.nba.com / sportsdataverse + ESPN
NCAAB	tempo-free (eFG, TO%, ORB%, FT rate), tempo, experience, KenPom/Barttorvik when posted	sportsdataverse + ESPN
MLB	Statcast EV/LA/barrel/whiff/chase, pitch mix, Savant BvP, order spot	Baseball Savant + MLB Stats API + ESPN
NHL	5v5 xG, GSAx, Fenwick, shift/line, goalie confirmed	MoneyPuck / NST / NHL API + ESPN
All	injuries, probable, goalie, starter, lineup	ESPN summary + league status feeds
All	weather / roof / wind / precip / temp	ESPN + forecast grid (NWS / Open-Meteo) labeled
All	rest, travel, timezone, altitude	live schedule log
All	crowd tape	Action Network tickets/handle, Kalshi, Polymarket

Host rule for ESPN remains: `site.web.api.espn.com`. `Number(object)` is still banned. Always `parseEspinScore`.

3.3 Tape / crowd

Unchanged reconstruction law from `tape.ts`. Handle split ≥ 7 points is a sharp tell, never auto-followed. ESPN win% is not handle.

4. Latent game state `G`

Function: buildLatents. Hidden vector, not a single probability.

Key	Meaning
muH, muA	expected home / away score (or goals/runs)
pace	possessions / plays / PA per game
sigM, sigT	margin and total σ
style	pass rate, ground-ball rate, 3-rate, shot-share, etc.
pWinH	P(home wins the graded market — 2-way including OT unless 3-way)
pRegH, pOT, pSO	NHL / NCAAB-OT split when the ticket needs it
chaos	extra variance from high total, backup QB/goalie, weather, live

Construction:

1. Market family still posts a moneyline / spread / total look (\$4.1 of the old spec, precision table unchanged).
2. Process family overwrites μ and σ when Ran (EPA, xG, KenPom, Statcast).
3. Context family (form, rest, weather, park, injuries-as-count) nudges logits and σ .
4. Usage vector u (\$5) reallocates player means; team μ is the sum of player contributions plus noise, not an independent number. If they fight, team latent wins for sides/totals; player latent wins for that player's tickets after a 30% pull toward the team residual.
5. Chaos: $\text{clip}((\text{total} - \text{league}) / (\text{league} \cdot 4), 0, 0.22)$ plus backup-QB +0.08, backup-goalie +0.06, wind $\geq$ 20 mph +0.05, outdoor precip +0.03.

pWinH is the **sim share of home wins** when the simulator Ran, else the pooled logit. Both numbers are stored.

4.1 Process looks that must always be attempted

NFL / NCAAF

- Offensive and defensive EPA/play, success rate, explosive rate (old 10 last games + season 40/60).
- Passing EPA vs rush EPA. Backup QB is a discrete state: if starter out, swap in backup EPA prior (career / this year) and widen $\sigma \times 1.18$.
- OL vs front as a rush-EPA / pressure residual when posted; else Looked.
- Red-zone TD rate separate from yards. Anytime-TD λ uses RZ + opportunity, not yards $\times$ constant.
- College: SP+ / FPI / explosiveness / returning production when posted. November last-3 is Thin.

NBA / NCAAB

- Four factors + tempo. Pace is possessions, not total / 224.
- Lineup net rating when the starting five is posted.
- Shot quality vs finishing for player points.
- College: KenPom/Barttorvik tempo-free when posted; experience and SOS. Transfer/portal Looked if the file is empty.

MLB

- wOBA-ish $0BP \times 0.9 + ISO \times 0.7$ remains the cheap look.
- Statcast barrel / whiff / chase overwrite when Ran.
- Starter 61% + bullpen 39% run-rate mix; bullpen B2B ± 0.08 logit on the game; pitcher K/9 $\times$ expected IP for K tickets.
- Order spot $\rightarrow$ expected PA (leadoff 4.4, middle 4.15, bottom 3.6, bench 1.8) then $\times$ platoon and pitcher.

NHL

- 5v5 xG for / against. Goalie GSAX. Confirmed starter vs backup is a discrete state (backup $\times 0.88$ team win logit bump to the opponent, $\sigma \times 1.12$).
- Special-teams xG as a separate slice for first-goal and period tickets.
- Puck line is **not** a win proxy. Same law as MLB run line.

4.2 Context family (always attempted)

Keep the old stamps and precisions: form, margin, trend, h2h, venue-split, series, injuries-count, rest, weather, park.

Weather law, sharpened:

- Indoor (NBA / NHL / NCAAB / dome): no ML lean from weather. Totals / air-game props still Looked-thin with the reading printed.
- **MLB:** weather moves totals, HR, and Ks. We still **do not flip an ML on wind**.
- **NFL / NCAAF:** wind is a **passing-EPA and total** modifier, not a $+0.02$ ML party trick.
 - wind 12-19: pass mean $\times 0.96$, rush $\times 1.03$, total -1.0 .
 - wind ≥ 20 : pass $\times 0.88$, rush $\times 1.08$, total -2.5 , deep-TD $\lambda \times 0.70$.
 - temp $\leq 32^\circ\text{F}$: $+0.04$ home logit (travel + crowd), rush + small.
 - precip $\geq 40\%$: pass $\times 0.94$, fumble chaos $+0.03 \sigma$.
 - $\geq 95^\circ\text{F}$: -0.015 home logit.

Park law: MLB event-specific factors when the table exists (HR, 1B, K). Fallback remains the run-factor list from 2026.09.08-live.

Injury **count** layer stays (precision light). The real work is §5.

5. Lineup / usage vector `U` (every player ticket, and every game ticket)

Function: buildUsage.

This is the layer the old desk faked with outs $\cdot 0.038$.

1. Start with each player's season + last-10 opportunity: snap %, route %, target share, rush share, USG%, ice time, batting order, line/pair.
2. Apply confirmed lineup. Out / IL / doubtful / suspended $\rightarrow$ opportunity 0, **stand down that player's tickets**.
3. Transmit leftover opportunity to remaining players with a **depth chart kernel**:

- NBA: extra USG and minutes go first to the same-position rotation, then to the usage magnet (star). Starter out $\rightarrow$ bench minutes $+6$ to $+12$ prior, clipped by that player's career minutes ceiling.
- NFL: WR1 out $\rightarrow$ WR2/slot target share $+0.08$ / $+0.05$ prior; TE/RB dump $+0.03$. RB1 out $\rightarrow$ RB2 rush share inherits 0.70 of the lost share, committee the rest. QB out $\rightarrow$ whole pass-tree rebuilds.
- MLB: scratch in the order $\rightarrow$ everyone below moves up one PA slot.
- NHL: line mate out $\rightarrow$ that line's remaining ice time is not 1-for-1; 2nd line bumps.

4. Questionable / game-time decision: run **two** usage vectors (in / out) and mix 0.55 in + 0.45 out unless the desk has a more specific participation look. Tickets on that player itself stay Thin and cannot be The Call.

5. College player usage runs **only** to move G (pace, mu, style). No college player ticket is emitted in Florida.

Precision: **5.8** when the starting lineup / batting order / goalie is confirmed; **2.4** projected; **0.9** Looked.

6. Player tickets

Function: buildPropChance. Hidden number: **P(over)** or **P(yes)**. Photographed side maps to chance-to-hit.

Always-pushed looks (old 6.1) still run: market 14, last10, total/pace, script, weather, park, rest, defense-allowed, usage, pitcher, underlying, platoon.

Additions that always run:

- **Opportunity (u)** — \$5. Precision 5.8 / 2.4.
- **Matchup process** — Statcast BvP, EPA vs position, 5v5 xG vs line, shot-quality vs defender. Precision 3.6 Ran / 1.4 Thin.
- **Script v2** — from sim win distribution, not a point P(home):
 - pass/rec yards rise when the team's path-share of trailing states is high;
 - rush yards rise when leading-state share is high;
 - anytime TD uses RZ trip distribution from the same paths.
- **Alt lines** — priced from the sim CDF, not by sliding one σ . Quality haircut -0.04 per σ of alt distance beyond 1.0.
- **Combos** (PRA, HRR, rush+rec): sum the path-wise components. Do not add independent Poissons.

Florida college player = illegal_fl stand-down immediately.

Listed-out / IL / doubtful-and-ruled-out / suspended = omit cells.

Sheet λ models from 2026.09.08-live §6.3 remain the **closed-form prior** inside each path. Process looks overwrite λ when Ran.

7. Live state machine `S`

Pre-tip s is empty. The moment the league clock / first pitch / puck drop is official, every open ticket on that event is tagged in_play and re-priced from S.

7.1 State vector

Sport	S
NFL / NCAAF	score, QTR, clock, down, distance, yard line, possession, timeouts, score phase, injuries since kickoff, weather now
NBA / NCAAB	score, QTR/half, clock, possession, bonus, foul trouble (4/5/6), timeouts, on-court lineup if posted
MLB	inning, half, outs, bases, score, pitcher pitch count, rest of lineup due up, bullpen already burned
NHL	score, period, clock, strength (5v5/PP/PK), goalie in net, empty-net flag

Source: ESPN summary / league gamefeed. Empty field = Looked and the live ticket **cannot** be ribbon / The Call. Microbets with a missing critical field (down, outs, strength) **stand down**.

7.2 Conditioning

```
paths ← paths that are statistically consistent with S
      or, cheaper and default: restart the remaining-game generator from S
P_live = mean(remaining paths hit the proposition)
```

Remaining-stat props (QB passing yards live, pitcher Ks live, points already scored):

```
remaining_mean = E[final | S] - already
P(over line) = P(already + remaining > line | S)
```

A player who has 62 rushing yards at 1:12 Q4 with the team kneeling is not a 72-yard prior.

7.3 Live quality

- Base live multiplier **0.52** remains the floor for ranking quality.
- If s is complete and process pbp on this game has Ran for ≥ 8 minutes (NBA), 3 drives (NFL), 3 innings (MLB), or 8 minutes (NHL): $\text{quality} \max(0.52, 0.52 + 0.20 \cdot \text{completeness})$.
- Still never The Call. Still never a ribbon parlay leg.

No generated 1st-inning sheet cells after first pitch. Posted live rows still rank on Live.

8. Ticket registry — every Hard Rock Florida family

If Hard Rock posts a market the registry does not know: stamp `unknown_market`, **stand down**, log the raw name. Do not silently price it as a cousin.

`infoQuality` is the ranking quality. The Call floor stays **0.72**. Live multiplies after the table.

Family	How it is priced	Quality
Full-game moneyline (2-way)	sim win share	1.00
NHL 3-way regulation	sim regulation share	0.90
Spread / run line / puck line	sim margin vs line	$0.94 \pm \text{key-number tilt}$
Alternate spread	sim margin tail	$0.86 - 0.04 \text{ per extra } \sigma$
Game total	sim total vs line	0.86
Alternate total	sim total tail	$0.80 - 0.04 \text{ per extra } \sigma$
Team total	sim team score	0.70
Counting player prop	sim player stat	0.80
Alt player prop	sim tail	$0.72 - 0.04 \text{ per extra } \sigma$
Anytime / 2+ / HR / SB / goal	sim counting process	0.68
First / last scorer	order statistic on scoring paths	0.58
Next scorer (live)	order statistic from S	$0.50 \times \text{live}$
Race-to-X / first basket	opening-possession model + sim	0.55
Odd/even, highest quarter	sim path functional	0.48

Family	How it is priced	Quality
Correct / exact score	sim mass on that cell	0.40
OT yes/no	sim OT flag	0.62
F5 / first 3 / first 7	sim truncated innings	0.64
Quarter / half / period winner	sim slice + shrink	0.50–0.58
Period total / period team total	sim slice	0.46–0.58
1st-inning 0.5	sim first inning	0.36
Current drive / next play / next pitch	micro generator from S	0.34 × live
Teaser	sim margin vs teaser chart (3 and 7 physics in NFL/NCAAF)	0.60
Boost	underlying quality, juice flagged	underlying
Future / championship / season prop	season-sim or market-only blend	0.58
Same-game parlay	joint sim of legs	0.62
SGP Max / cross-game	joint sim; independent across games after shared weather/public residual	0.55 / 0.40
Flex / consolation / round robin	price each included combo, show the set	combo quality
Floor	—	0.28
In-play multiplier	applied after the row	× 0.52 (see §7.3)

NFL/NCAAF key numbers 3 and 7 stay as ranking tilt **and** now sit inside the sim: extra mass on margins of 3 and 7 via a discrete spike of 4.5% of path weight split across {3, −3, 7, −7} when both teams are NFL/FBS. Do not fake cover% another way.

Period shrink from the old table is the **closed-form look** that still goes into the pool. The sim slice is the primary when it Ran.

9. Parlays

9.1 Joint probability

Default: **the legs ride the same paths.**

```
raw = mean( all legs hit on path i )
```

Cross-game: paths are independent **except** a shared residual ρ when the events share weather system, same-city back-to-back, or a slate-wide public blowout. Desk $\rho = 0.04$ default, 0.10 if the same weather grid cell is Ran for both outdoor games. Else 0.

Fallback if a leg cannot be emitted from the path engine (Looked generative model): old haircut table, labeled Thin.

Legs	Includes ML+spread	Haircut fallback
2	yes	0.55
3+	yes	0.42

Legs	Includes ML+spread	Haircut fallback
2	no	0.88
3	no	0.75
4	no	0.62
5-8	no	0.50
9+	no	0.38

Same-game ML+spread must use the sim, not the haircut, whenever both legs are registered.

```
combined = min(0.97, raw)
```

9.2 Ribbon law (AI-picked, paramount)

Goal: **highest honest hit chance** among tickets the desk still trusts. Not lottery posters.

Must be all of:

- 2 or 3 legs. 4+ live on the Parlay tab only. Never the gold badge.
- No in-play legs.
- No college player legs.
- No player props on the **badge** (they still rank as singles and as custom legs). Ribbon is main-market: ML / spread / total / team total / F5.
- Each leg displayed chance $\geq 56\%$ (2-leg) or $\geq 60\%$ (3-leg).
- Each leg `infoQuality` ≥ 0.72 .
- Combined displayed $\geq 38\%$ (2-leg) or $\geq 28\%$ (3-leg).
- Combined decimal payout ≥ 1.45 .
- Score > -90 .

Sort ribbon by `combinedFair` desc, then `parlayScore`, then `id`. This is the high-percentage guide.

Propose up to **6** ribbon slips. Cap 1 slip per event on the badge.

9.3 Custom builder (user freedom)

Any legal Florida combination the registry knows. The page prints, live, per added leg:

- leg fair, HR implied, edge
- running joint fair and running decimal
- correlation flag: `shared-latent` / `near-independent` / `fallback-haircut`
- stand-down the whole slip if any leg is `illegal_fl`, `unknown_market`, or live-micro with incomplete S

Catalog floors (not badge): 2-leg 48%, 3-leg 50%, 4-leg 48% per leg before haircut.

9.4 Score

```
parlayScore = combinedFair^1.6 · ln(1 + profit) · 8 + combinedFair · 2.4
```

The exponent on fair is higher than 2026.09.08-live on purpose: hittable 2-legs beat 17-leg posters.

Then $\times$ `infoQuality` (0.70 / 0.55 / 0.40 / 0.62 for 2 / 3 / 4 / SGP).

Typical parlay juice flag only: 12% (2) / 25% (3) / 32% (4) / 40% (5+). EV $< -8\% \rightarrow$ titled “fun money.”

9.5 Teasers / flex / round robins / SGP Max

- Teaser: shift each NFL/NCAAF spread by the chart (6 / 6.5 / 7) and re-score the **same paths**. Key-number crossings are the whole point.
- Flex / consolation: print $P(\text{all})$, $P(n-1)$, $P(n-2)$ from the same paths; do not fake with binomial independence.
- Round robin: expand to the combo set, cap display at 12 combos, link the rest.
- SGP Max: same-game blocks use joint sim; blocks multiply with the tiny cross-game p .

10. Ranking, calibration, The Call, mood

10.1 Calibrated displayed chance

```
implied = implied(Hard Rock FL two-way) # photo or hardrockbet_fl
displayed = implied + (fair - implied) · (0.35 + 0.65 · quality)
```

No Florida implied: $0.5 + (\text{fair} - 0.5) \cdot (0.4 + 0.6 \cdot \text{quality})$.

Clip at 99%.

10.2 Edge

```
edge = fair - implied(Hard Rock FL)
```

If only a public two-way exists, print **Research edge** and refuse to title it a Florida fill.

10.3 deskScore / ticketScore

Unchanged algebra from 2026.09.08-live, including the big-favorite haircut so -400 at 80% loses to -110 at 58%, and the tape bump (sharp $+0.055$, public -0.035).

```
ticketScore = deskScore · quality + clip(edge · 1.15, -0.07, +0.14)
```

Tie-break: score, then chance, then id. No free dice.

10.4 Confidence badge

- **High:** quality ≥ 0.88 , chance ≥ 0.52 , edge ≥ -2.5 pts, sim and pool agree within 4 pts.
- **Medium:** quality ≥ 0.68 , chance ≥ 0.48 .
- **Low:** otherwise.

10.5 The Call (gold badge)

Must be **all** of: a single; not in-play; decimal payout ≥ 1.55 ; displayed chance in **[50%, 76%]**; infoQuality ≥ 0.72 ; score > -90 ; registry family is not a period slice, exact score, first-scorer, micro, future, or teaser.

Fallback (still never live, never a parlay): payout ≥ 1.55 , chance $\geq 48\%$, quality ≥ 0.64 .

10.6 Mood

Does **not** rewrite chance.

- **Safest (Chill)** — highest chance, then score, then id. Default for users who said they want to hit.
- **Best value (Skill)** — ticketScore.
- **Pays more (Thrill)** — biggest payout among tickets with quality ≥ 0.64 and chance ≥ 0.48 .

Ribbon parlays always sort as Safest on the badge, even if the singles desk is in Skill.

10.7 Catalog caps

Popular 8 · Player 8 · Periods 6 · SGP 6 · 2-leg 6 · 3-leg 6 · 4-leg 4 · Live 8.

Per-event cap: **1 period**, **2** otherwise, then fill.

Horizon: today ET or in-play or next 36 hours (and games that started in the last 4 hours).

10.8 Legal / stand-down gates

- In-play → `in_play`. Ranks on Live only.
- DK/FD sportsbook venue as the fill → `illegal_fl`.
- College player prop → `illegal_fl`.
- Unknown registry name → `unknown_market`.
- Live micro with incomplete S → stand down.
- Fair or better vs Hard Rock two-way ($EV \geq 0$) → `fair_or_better`.
- Main market, $EV \geq -3\%$ → `close_enough`.
- Else juiced / missing two-way → stand down on the **scan** tag. AI Picks may still rank the research look; the ticket page says photograph Hard Rock (or wait for `hardrockbet_fl`).
- One-sided market with no model hold → cannot call a fair two-way. Stand down on that scan row.
- Seed bankroll < \$100: still **names** a ticket. `liveFits` is false until bankroll $\geq$ \$100 and unit $\geq$ \$1. Naming is not placing.

Failure-mode law (the one rule):

Situation	Action
Illegal in Florida	stand down
Unknown market	stand down
Live micro, critical S missing	stand down
Player listed out	omit
Optional look empty	Looked, keep going
Small sample	Thin, keep going
Generative model empty, closed form exists	use closed form, quality cap 0.64
Both generative and closed form empty	Looked 50/50, quality floor 0.28, never badge

11. Futures

Season-sim when the slate of remaining games is loadable: replay each remaining game with week-updated G (injuries persist, variance widens as week count drops). Else market-only: Hard Rock future no-vig if two-way / field can be rebuilt; else Looked.

Quality 0.58. Never The Call. Never a ribbon leg.

12. Official rule list (ticket page stamps)

Same list every time. `rules.ts` ALGORITHM_RULES. Stamp via `ruleStamp`.

- 1. Hard Rock Florida is the fill** — photographed ticket or Ran `hardrockbet_fl`. Public books and prediction markets are research. Layers: market, book, open, spread, hr-fl.
- 2. Latent game, not one number** — win, pace, variance, style. Sides and totals are not transforms of each other. Layers: latent, efficiency, pythag, espn.
- 3. Seeded paths** — every registered ticket on an event rides the same paths. Alts, first-scorer, parlays, live remaining. Layers: sim.
- 4. Last 10 — every stat, not just W-L** — EWMA. Periods inherit then shrink. Layers: form, margin, trend, last10, season-rate.
- 5. Lineup transmission** — scratches move usage, order, ice time. Count-of-bodies is the light backup. Layers: usage, injuries.
- 6. Process looks** — EPA / four factors / Statcast / xG / tempo-free. Empty = Looked. Layers: process, underlying, defense, platoon, pitcher.
- 7. Head-to-head and venue** — last-10 meetings → vs-opp → series; last-10 home vs road. Layers: h2h, venue-split, series.
- 8. Open → close and tape** — steam with the handle is informed money, still not a lock. Never copy 80% of tickets. Layers: steam, open, tickets, handle.
- 9. Kalshi + Polymarket** — a different crowd. Research only. Layers: kalshi, polymarket.
- 10. Rest / B2B / bye / travel** — NBA/NHL B2B hurts. NFL bye vs short week. MLB extra rest is small except bullpen B2B. Layers: rest.
- 11. Weather and park** — wind/rain/heat move totals and the air game more than moneylines. We do not flip a baseball ML on wind. Coors is not Petco. Layers: weather, park, total.
- 12. Live is a state machine** — score, clock, downs, outs, strength. Remaining props use leftover mean. Quality floor. Never The Call. Layers: live-state.
- 13. Period markets are noisier** — sim slice + shrink toward 50/50. Quality floor 0.72 so they cannot be The Call.
- 14. Parlays are joint paths** — same-game correlation is simulated. Cross-game is almost independent. Ribbon maximizes hit chance, not poster odds.
- 15. Calibrate thin markets** — displayed chance is pulled toward Hard Rock when info-quality is low.
- 16. Edge vs Florida juice** — rank chance² × √payout × quality + edge. 58% at -110 beats 80% at -400.
- 17. CLV + calibration loop** — layers that lose CLV get quieter. Miscalibrated badges get pulled toward the book.
- 18. Florida law** — college player bets blocked. No scraping Hard Rock. Photograph when the tape is missing or boosted. This site never places a bet.

Rules with no layer id stamp **Ran** whenever that ticket type is shown.

13. Determinism

- No unseeded `Math.random` in `src/lib/market`.
- Pure functions: same snapshot + same `DESK_VERSION` → same latents, same paths, same chance, same `ticketScore`, same hero id, same ribbon.
- Sort: score, then chance, then id.
- `DESK_VERSION 2026.09.09-world`. Bump when a ranking rule changes.
- Fingerprint: id | chance(6 dp) | score(6 dp) | bucket | seed.
- Tests pin: same 10 scores → identical `FormRead`; empty feed → Looked not skip; same snapshot → identical path hash; leftover four layers exist on game and prop reports; college player ticket → `illegal_fl`; unknown market → stand down; live without downs/outs/strength → micro stand-down.

14. What is never done

- Never invent a missing split, EPA, xG, Statcast, minutes, last-10, or handle %.
- Never print NO LOOK. Empty = Looked.
- Never scrape Hard Rock or invent its order book.
- Never treat ESPN `lastFiveGames` as analysis.
- Never treat last-10 **hit rate** (8/10 cashed) as a lock.
- Never auto-copy 80% public.
- Never flip a baseball moneyline on wind.
- Never add home-field twice on top of a home/road split.
- Never use own points-per-game as “defense.”
- Never emit a Florida college player ticket.
- Never let a period 0.5, a live ticket, a future, a teaser, an exact score, or a 4-leg be The Call.
- Never treat a boost as edge.
- Never place a bet. Never call a lock. Never print 100%.

15. Source map (audit)

Claim	Where it should run	Published / desk
Logit pool + overdispersion + JS pull	<code>chance.ts poolLayers</code>	published method; 1.35 / λ desk
Seeded PCG64 paths	<code>sim.ts drawPaths</code>	desk; determinism law
Abramowitz Φ	<code>chance.ts normalCdf</code>	A&S 7.1.26
$\log 5$ / Pythagorean	<code>chance.ts</code>	James / Oliver / FO
EWMA $\lambda=0.82$, 60/40	<code>form.ts</code> , <code>props.ts</code>	StatPair / AgentBets; λ desk
Hard Rock FL tape	<code>tape.ts hrFL</code>	The Odds API <code>hardrockbet_fl</code>
Photograph overrides feed	<code>lock.ts</code>	user law
NFL EPA / CPOE / xpass	<code>process.nfl.ts</code>	nflverse / nflfastR
College SP+ / tempo-free	<code>process.cfb.ts</code> , <code>process.cbb.ts</code>	cfbfastR / KenPom-family when posted
NBA four factors / usage	<code>process.nba.ts</code>	stats.nba / sportsdataverse
Statcast / Savant	<code>process.mlb.ts</code>	Baseball Savant
NHL xG / GSAx	<code>process.nhl.ts</code>	MoneyPuck / NST / NHL API
Usage transmission	<code>usage.ts</code>	desk kernel
Live state machine	<code>live.ts</code>	desk
Key numbers 3 and 7	<code>sim.ts</code> + <code>picks.ts</code>	Dimers / CLV / NFL physics
Joint parlay paths	<code>parlays.ts</code>	desk; haircut is fallback
Calibrate toward Florida tape	<code>picks.ts calibratedChance</code>	Olympus / CLV thin-market
CLV precision loop	<code>calibrate.ts</code>	desk
Florida college-prop block	<code>props.ts</code> , <code>engine.ts</code> , <code>universe.ts</code>	Florida compact / Hard Rock Bet
Always-push empty layers	<code>chance.ts</code> / <code>props.ts pushEmpty</code>	user law

Claim	Where it should run	Published / desk
Rule stamps Ran / Thin / Looked	rules.ts ruleStamp	user law

Companion: `SPORTS_LOCK_BLUEPRINT.md`. User prompts in chat override both when they conflict. Append a dated note when a ranking rule changes, and bump `DESK_VERSION`.

Picking with Intelligence Creates Confidence.

Not a lock. Photograph Hard Rock to lock the live number when the Florida tape is missing. This site never places a bet. 1-800-GAMBLER.